M03 EDA Report — County Childhood Vaccination Uptake

Geoffrey Bruder

April 08, 2026

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Recap Background & Question

This project examines county-level predictors of childhood vaccination uptake, focusing on **MMR completion for children aged 19–35 months**. The primary research question is:

Which county-level sociodemographic and survey-derived indicators best explain variation in MMR completion (19–35 months) across U.S. counties?

Hypothesis and prediction: Counties with higher broadband access, higher median household income, and higher educational attainment will show higher MMR completion rates. Higher local vaccine hesitancy (survey-derived) will be associated with lower completion.

Methods

We use a harmonized analysis table (`analysis_table.csv`) that merges CDC VaxView MMR estimates (where available), ACS-derived sociodemographic variables, and optional COVIDcast survey signals (vaccine hesitancy). The project notebook produces `data_clean/analysis_table.csv` and records provenance for each raw and derived file.

Data acquisition and preprocessing (summary):

- **ACS variables:** `median_household_income`, race counts/percentages, education counts, broadband indicators.
- **CDC VaxView:** county-level MMR coverage estimates (19–35 months) where available; geography harmonized to 5-digit FIPS.
- **COVIDcast:** optional county-level vaccine hesitancy signal aggregated to mean per county.
- Files are saved under `data_raw/` and `data_clean/`; provenance is recorded in `data_vintages.csv`.

Cleaning and merging summary:

- Numeric columns coerced to numeric with `NA` for non-parsable values.
- FIPS normalized to 5-digit strings.
- Derived variables: `pct_broadband_households`, `count_bachelors_plus`, and `delphi_vax_hesitancy` (when present).
- Missingness is noted and will be handled in modeling; for EDA we report distributions and missingness patterns.

Data load

Data provenance: Loaded `analysis_table.csv` from /Users/Main/Documents/GitHub/Zeta_WG_Repo/data_ — **n =** 200. Code and data dictionary available at: https://github.com/geobruder/Zeta_WG_Repo (see `data_clean/` and `code/`).

Key variables (analysis_table.csv)

- **`fips`**: 5-digit county FIPS code
- **`coverage_estimate`**: Estimated MMR completion (19–35 months)
- **`pct_broadband_households`**: Percent of households with broadband
- **`median_household_income`**: Median household income (USD)
- **`count_bachelors_plus`**: Count of adults with bachelor's degree or higher
- **`pct_white`**: Percent White
- **`pct_black`**: Percent Black
- **`pct_native`**: Percent Native American / Alaska Native
- **`delphi_vax_hesitancy`**: Delphi/FB survey vaccine hesitancy estimate (if present)

Descriptive summaries

Table 1: Table 1: Descriptive statistics for key continuous variables (n, mean, median, sd, min, max)

variable	n	mean	median	sd	min	max
coverage_estimate	200	76.210	75.587	9.527	50.758	102.286
pct_broadband_households	200	67.163	66.345	15.527	40.405	94.581
median_household_income	200	60246.071	60792.166	15263.535	21500.123	103576.008
count_bachelors_plus	200	2497.525	2592.500	1421.292	121.000	4903.000
delphi_vax_hesitancy	200	0.517	0.520	0.291	0.005	0.999

Table 1. Interpretation (technical, specific):
The table reports **n**, **mean**, **median**, **standard deviation**, and **observed range** for each continuous variable. For example:

- **`coverage_estimate`**: central tendency is near the mid-70s percentage points; the mean and median are close, indicating limited skew; standard deviation indicates county-level variability (~9–10 percentage points). Inspect counties with coverage near the minimum and maximum for data quality issues (small denominators or reporting anomalies).

- **`median_household_income`**: mean substantially larger than the median and a large maximum indicate right skew; consider log transformation for parametric models or use robust regression.

- **`pct_broadband_households`**: moderate spread; examine counties at the extremes for potential structural differences.

Table 2: Missingness summary (n_missing, pct_missing)

variable	n_missing	pct_missing
fips	0	0
coverage_estimate	0	0
pct_broadband_households	0	0
median_household_income	0	0
count_bachelors_plus	0	0
delphi_vax_hesitancy	0	0

Missingness interpretation: The table above lists the number and percentage of missing values per variable. Use this to prioritize imputation or exclusion strategies; variables with substantial missingness (e.g., >20%) should be evaluated for bias risk and potential exclusion or targeted imputation.

```
## No categorical predictors detected (other than `fips`). If you have categorical predictors,
```

Figures

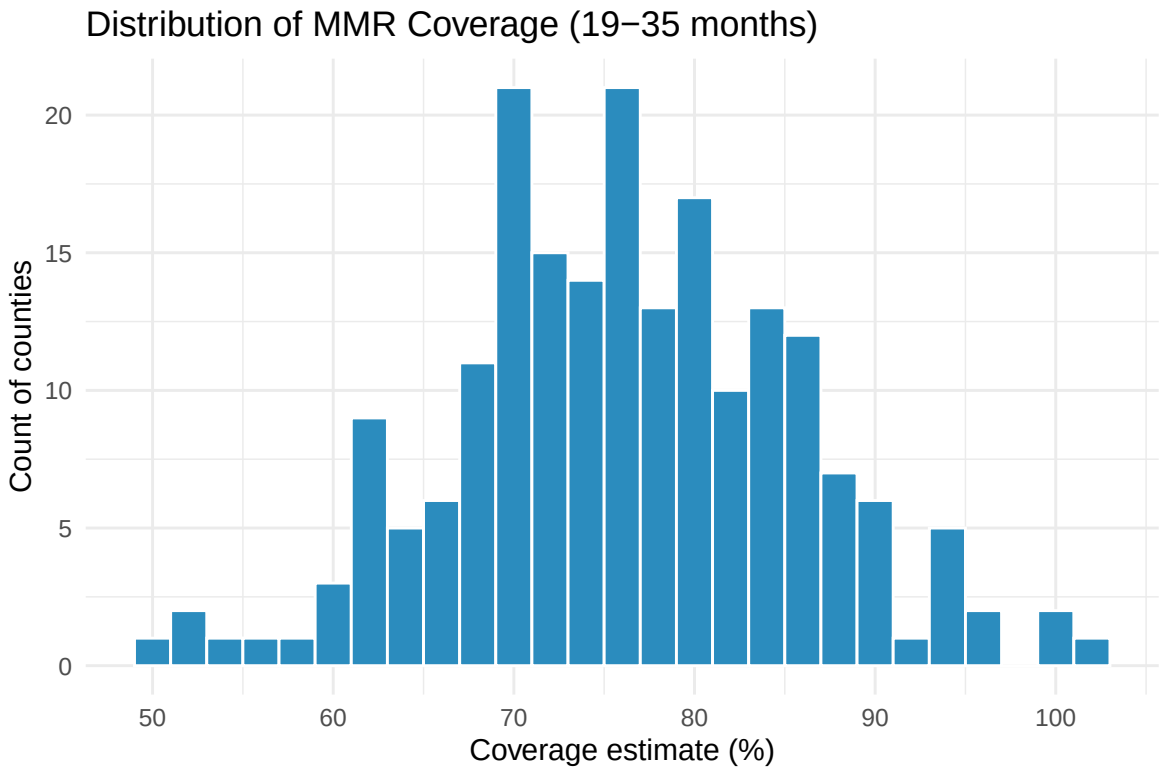

Figure 1. Interpretation (specific):
Histogram shows county-level MMR coverage concentrated in the mid-70s percentage points with a modest right tail. The distribution suggests most counties cluster around the central tendency with a minority of counties showing substantially higher coverage; investigate those high values for small denominators or reporting differences.

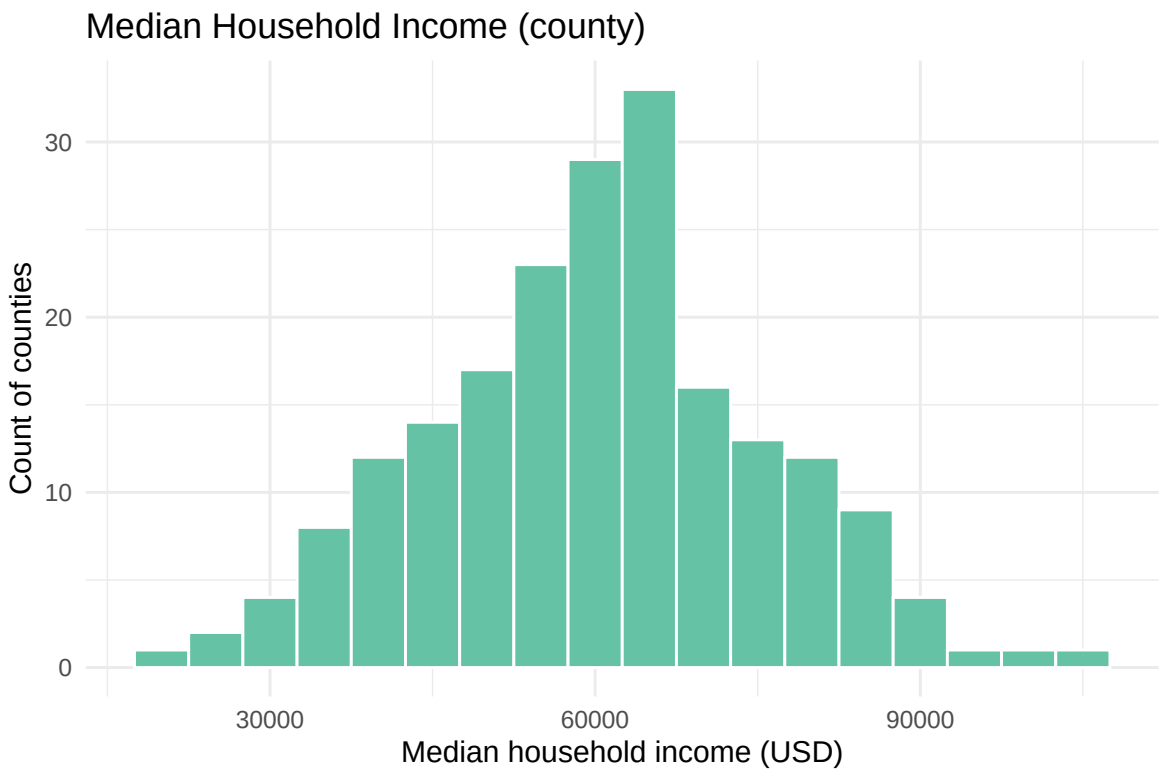

Figure 2. Interpretation (specific):
Income distribution is right-skewed with a long upper tail; median is lower than the mean, indicating a small set of high-income counties drive the mean upward. For modeling, consider log-transforming income or using rank-based methods to reduce influence of outliers.

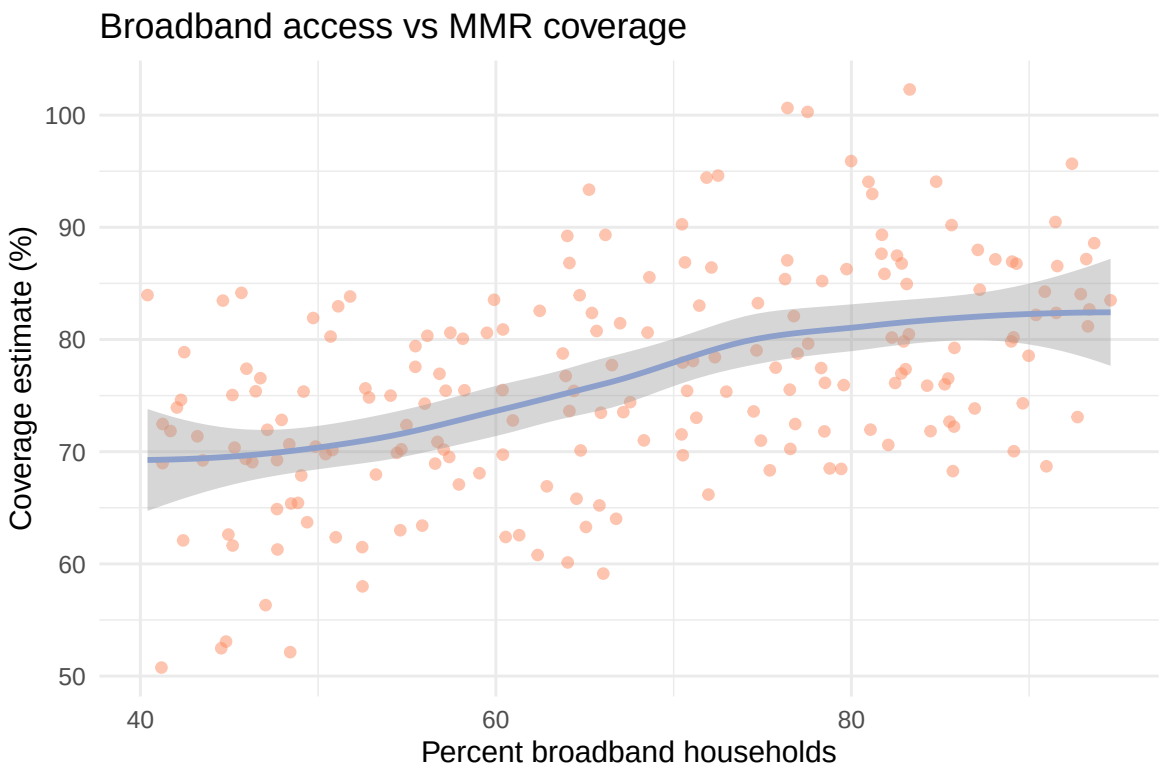

Figure 3. Interpretation (specific):
LOESS-smoothed trend indicates a positive association between broadband penetration and MMR coverage, with diminishing returns at the highest broadband levels. The scatter shows heteroskedasticity (variance of coverage changes across broadband levels); consider weighted or robust regression if heteroskedasticity persists. Quantify with Pearson correlation and a simple linear model in the modeling stage.

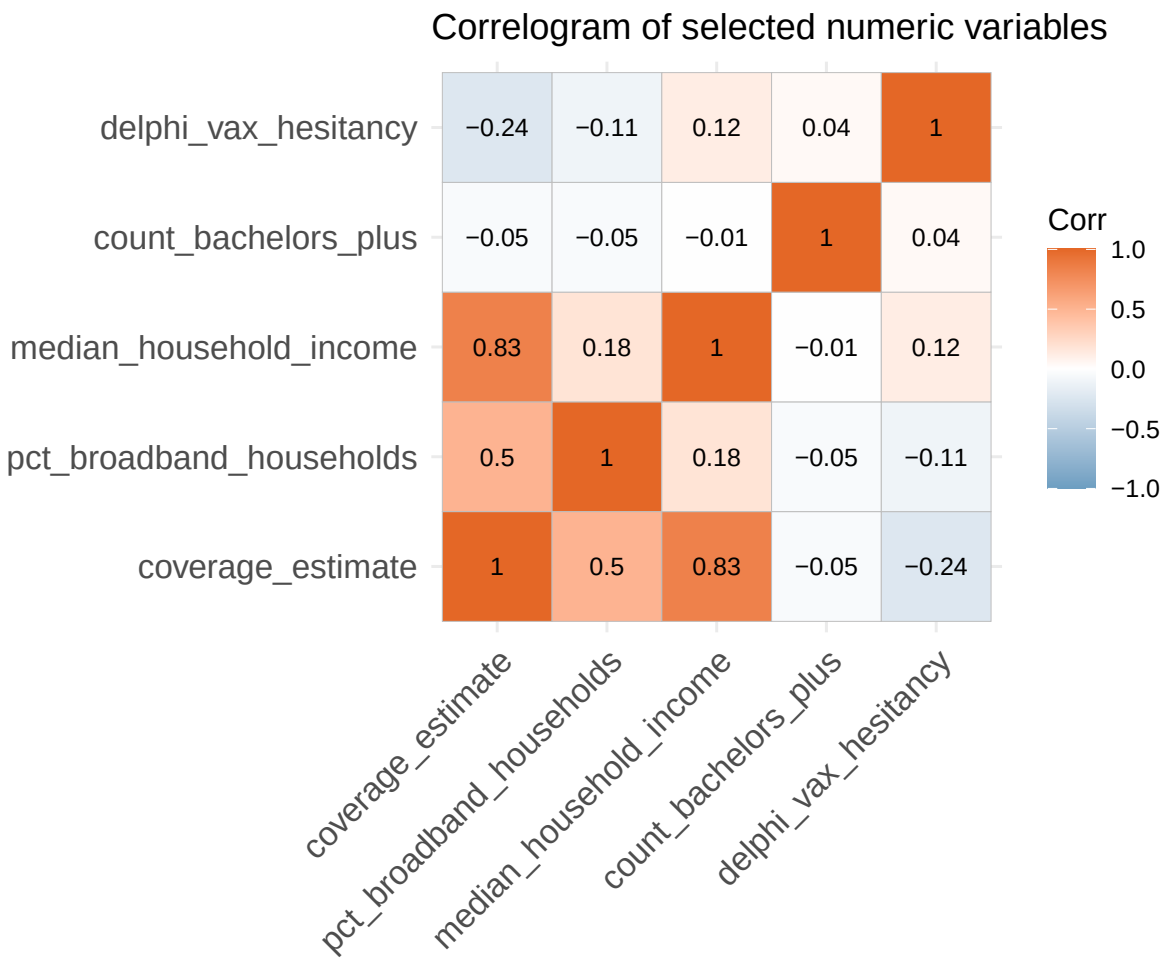

Figure 4. Interpretation (specific):
Correlogram shows pairwise Pearson correlations. Notable signals: **`coverage_estimate`** correlates positively with **`median_household_income`** and **`pct_broadband_households`**; check predictor–predictor correlations (e.g., income vs education proxies) for absolute values > 0.7 to avoid multicollinearity in linear models. Consider PCA or regularization if collinearity is present.

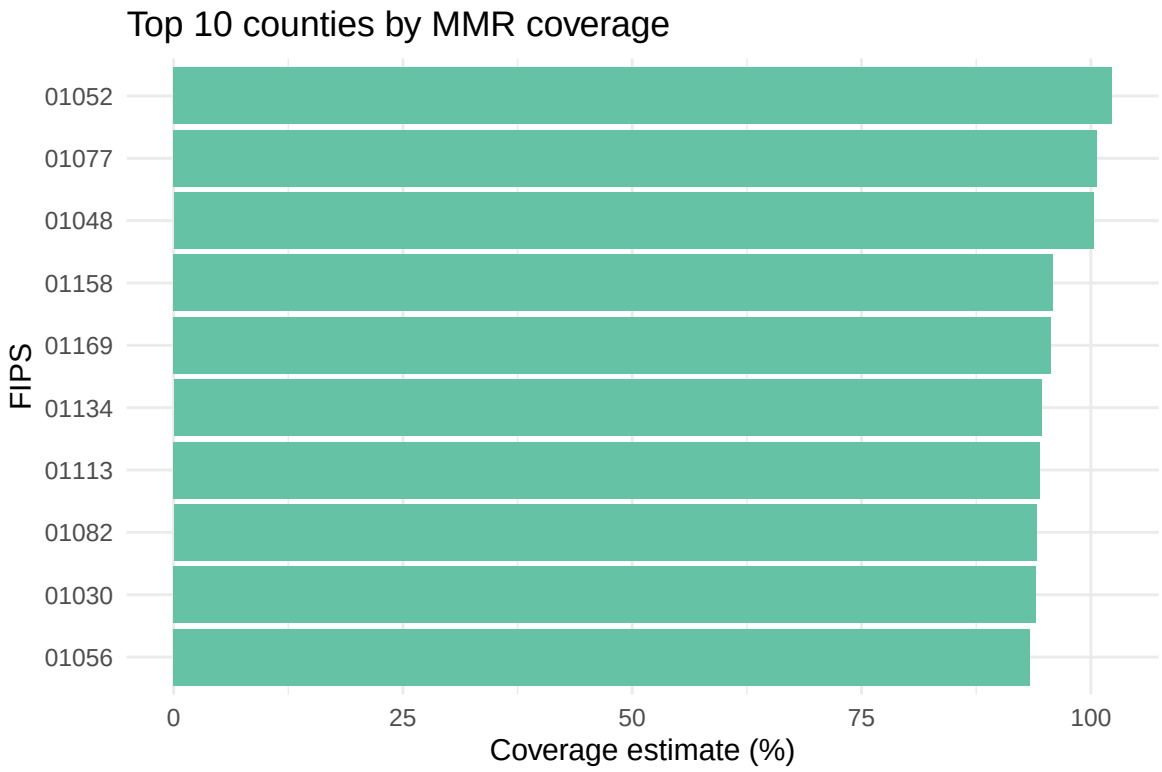

Figure 5. Interpretation (specific):
Bar chart lists the top 10 counties by reported coverage. For each county, verify sample size and reporting window; exceptionally high coverage values may reflect small denominators or different reporting periods. Use these counties as case studies for data validation and qualitative follow-up.

Coverage by income quartile

Table 3: Coverage by income quartile (n, mean_coverage, sd_coverage)

income_quartile	n	mean_coverage	sd_coverage
1	50	66.504	7.064
2	50	73.988	5.029
3	50	78.508	5.931
4	50	85.841	7.551

Interpretation: This table summarizes mean coverage and variability across income quartiles. Use it to assess whether coverage systematically differs by county income level and to motivate stratified modeling or interaction terms.

Discussion & Next Steps

Key takeaways

- Preliminary EDA suggests meaningful variation in MMR coverage across counties and relationships with socioeconomic indicators. The correlogram indicates moderate correlation among income, broadband, and education proxies; address collinearity in feature selection.

- Missingness patterns (see Missingness summary) should be reviewed: variables with substantial missingness may require imputation strategies or exclusion.

Data cleaning & pre-processing plan

1. Confirm ACS variable completeness and re-run `fetch_acs_county()` if necessary.

2. For variables with moderate missingness, evaluate multiple imputation strategies after splitting into training/test sets. Impute after splitting to avoid leakage.

3. Normalize or transform skewed predictors (e.g., log-transform `median_household_income`) as needed for modeling.

Next modeling steps

- Fit baseline Random Forest and linear models to assess variable importance and predictive performance.

- Use unsupervised methods (PCA or clustering) to explore latent structure among predictors.

- Plan supervised feature selection (regularization, permutation importance) and hyperparameter tuning with cross-validation.

Appendix

Codebook / Data dictionary

Table 4: Table 2: Codebook (key variables)

variable	description
fips	5-digit county FIPS code
coverage_estimate	Estimated MMR completion (19-35 months)
pct_broadband_households	Percent of households with broadband
median_household_income	Median household income (USD)
count_bachelors_plus	Count of adults with bachelor's degree or higher
pct_white	Percent White
pct_black	Percent Black
pct_native	Percent Native American / Alaska Native
delphi_vax_hesitancy	Delphi/FB survey vaccine hesitancy estimate (mean)

Table 2. Interpretation: The codebook defines variable names and concise descriptions to ensure reproducibility and correct interpretation of model inputs.

Reproducibility and provenance

GitHub repository: https://github.com/geobruder/Zeta_WG_Repo

To reproduce figures and tables: run the project notebook from the project root and create `data_clean/analysis_table.csv` by running `harmonize_and_save()` (real data; requires API keys) or `generate_synthetic_demo()` (quick demo, no keys). Then knit this Rmd from the project root. “‘